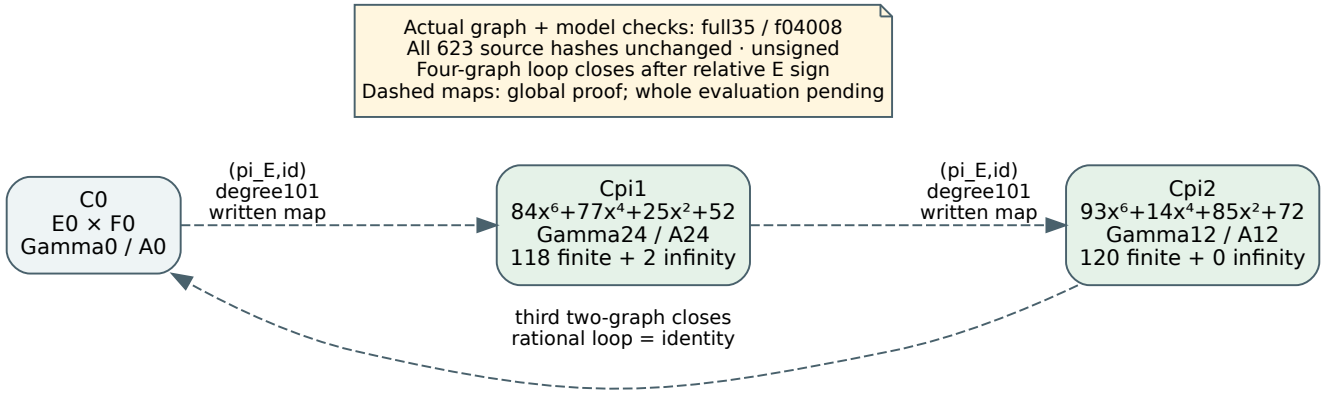
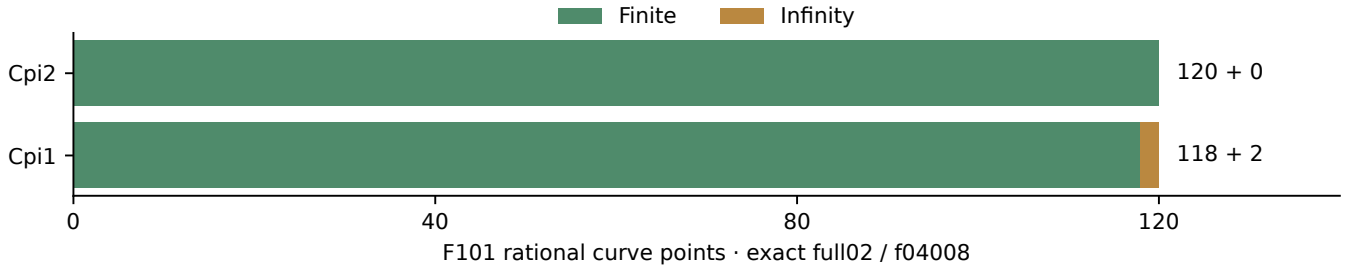
Actual partial Frobenius over F101

2026-11-12 · full35 · two explicit models · six bound local Lean proofs

The complete changed suite passes35 tests, zero failed/ignored/filtered; oldguards, formatting, strictClippy and all623 source hashes pass. Native full02/f04008 retains whole89,109-byte output as standardXZ10,120, with raw/archive hashes,source/status and exact roundtrip. Selected6bbfba is one diagnostic with34 filtered. All scientific standing is unsigned. [pcc:unverified]



Actual two-glue models and four-graph sign classes. All have factors E0=(1,46), F0=(44,21), orders109/113. Solid boxes are actual model/graph checks; dashed arrows are global written maps, not new whole Mumford evaluations. Cpi1/Cpi2 are distinct presentations inside the fixed72 catalog. [pcc:unverified]



Exact recorded counts, no sampling uncertainty. Both have N1=120,N2=10436, Jacobian order12317. Cpi1 has leading/constant roots±36/±31; Cpi2 has neither overF101. No retwist or monic normalization is applied. [pcc:unverified]

Actual geometric and rational controls

The source Q0 is extracted from elliptic4-product Cantor pulls. Each transport checks16 actual points,full projections,all256 sums/product Weil pairs and Galstability. One-factor Frobenius gives A24/A12, matching independently reconstructed balanced walks. Q3 equals the first-factor negative of Q0 exactly. The firstthree sign classes and two-graphs are distinct. The geometric identity mutant misses15/16 points in both cases. [pcc:unverified]

Both covers satisfy exact coefficient identities. Every rational curve point image, both extension infinities/x0 points, all6 branch roots and the actual two-graph pass. Direct/reciprocal parameters(destination,a,b/3,s) are(F0,84,93,9)/(E0,52,42,9) and(F0,93,72,37)/(E0,72,62,37). [pcc:unverified]

Frobenius fixes all109+113 rational factor points and all12317 product pairs for k1/k2. The signed three-step route equals CRT6103 on all12317 pairs;alternative sign6214. The two-graph third loop already has rational identity before the optional elliptic involution. [pcc:unverified]

Global maps and a direct-bridge comparison

Written proofd67ccf descends(phi_k)=(pi_E^k,id) through actual stable kernels. Degree is101^k, purely inseparable; the kernel scheme is connected of that length and has exactly one geometric point. No101-point kernel is enumerated. Adjoint norm descends diag(101^k,1), and degree coprimality preserves both109/113 DLP subgroups overF101. The AV/Picard/group-scheme bridges are written, not a full formal AV library. [pcc:unverified]

No additional factor-stage advantage over the direct bridge

With canonical pull/norm P_i,H_i and norm scalar n=2 or4, [n]F_k=P_k phi_k H0 globally. On rational points phi_k fixes H0(D), so F_k=[n]^-1 P_k H0. The existing direct bridge G=P_k H0 already has separable degree16/256 and preserves the odd-prime logarithm. Both have an identity factor stage;the common output multiplier n can be retained for DLP transfer. Partial Frobenius therefore supplies different global maps and actual target classes but no extra factor-evaluation saving over G. Construction,target pullbacks,solver and recovery remain unmeasured. [pcc:unverified]

After3 steps the four-graph sign identification gives rational(-1,1), CRT6103. The global degree101^3 map is not integer multiplication, whose degree on an abelian surface is m^4. Rational scalar agreement does not identify global morphisms. Explicit primitive-four target equations and whole Jacobian maps remain open. [pcc:unverified]

Six exactly bound local proofs

PartialFrobenius.frobenius_preserves_form/e9174e and partial_preserves_product_form/e19537 cover finite symplectic covariance; finite_sign_orbit/106caa covers $F^3=-I$,$F^6=I$ and no early sign. public_moduli_and_norms/72739d checks CRT and norm/inverse constants. public_inverse_halves_quarters/4b1b04 gives inverse2/4 in an IntModule killed by12317. rational_bridge_agreement/f99da7 proves route equality from explicit fixed-image,cleared-map and inverse hypotheses. Gate02 passes6/0trusted;all6 bindings locally pass. Project census177 written/35modules;203compiled is expected,not ownCI-observed. [pcc:unverified]

Actual torsion bases,Weil forms,varieties,quotients and degree bridges stay written/native. Old seven algebra certificates and two universal cover proofs remain unchanged;finite module lemmas supply no new polynomial certificate. Static336266/parser21c177/PATHa31d7b failures remain intact with a separate inspected report. e9174e’s earlier not_covered and new03 pass are both retained. No mathematical prediction failed. [pcc:unverified]

Sources, reproduction and next action

Primary:Galbraith9.6.14/15/24 and26.2.3;MilneAV I7/II1; Bruin–Doerksen0902.3480 Lemmas2.4/2.6/ Theorem3.2;existing outer-route CHAIN_COMPRESSION_PROOF.md. ExternalCT43cc5d5541 abstract orbit credited. KB MCP/wrapper/atlas unavailable;protocol e60790 records queries/fallback, with no novelty inference. Rust1.99.1,Lean4.35.0;focused workflow runs locked/offline release tests,guards/fmt/ strictClippy and all35 proof modules. REPORT.md links complete evidence,proofs and editable sources. [pcc:unverified]

Next:generic nonmonic chart and cover norms from input Mumford divisors; all12317 whole rational maps,duals/recovery and exception controls for Cpi1/Cpi2 and unbalanced C10/C20. New primitive-four targets/T20 and matched full-cost comparisons remain queued. No benchmark or paid run launched. [pcc:unverified]